



Unlearning is **recurring** in nature. Existing methods struggle with robustness due to unbounded losses, static trade-offs and over-forgetting.

BLADE: Repair-First Principle + Bilevel Optimization + Asymmetric ALM Ratchet + Clamped Entropy, confined to LoRA.

+6% TOFU, +9% MUSE Books, +7% KnowUndo; stable under 4× scale and 4 sequential steps.

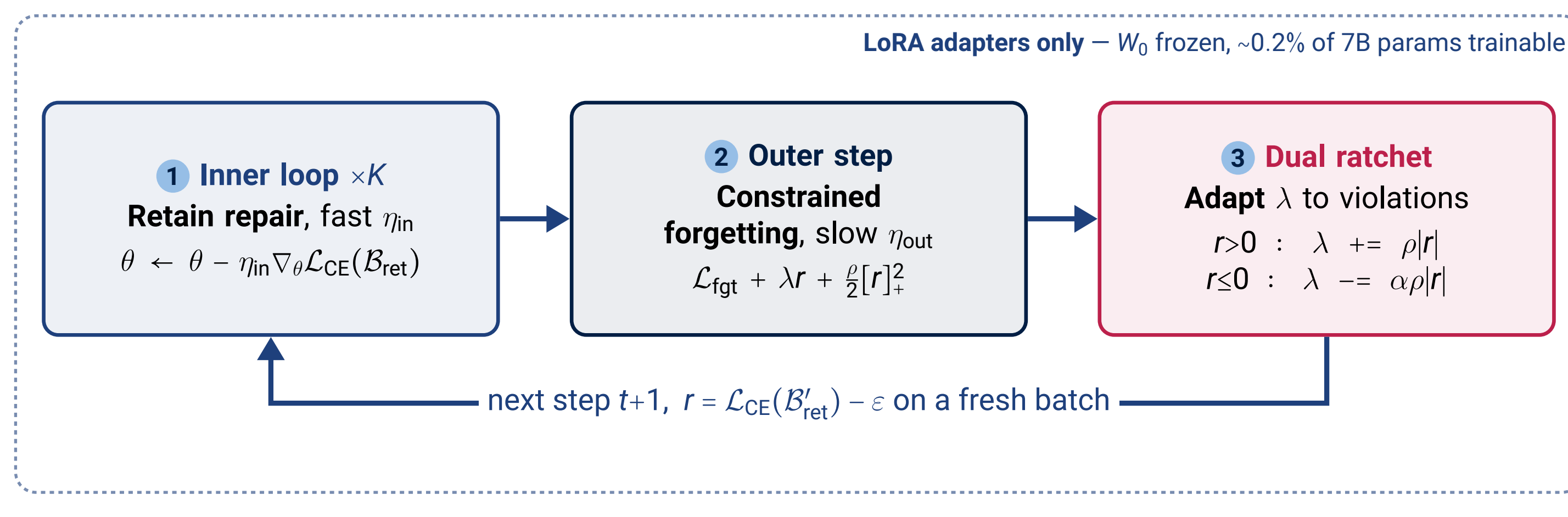
Why is LLM unlearning hard?

- Deployed LLMs need continuous correction: GDPR deletion requests, copyrighted text, hazardous knowledge, misinformation.
- Forget and retain knowledge share **entangled representations**: every step risks collateral damage that compounds.
- Fixed-weight forget/retain balancing yields three failure modes:
 - Unbounded forgetting**: gradient ascent³ has arbitrarily large gradients, destroying utility.
 - Static trade-off**: GradDiff, NPO¹⁰ oscillate without self-correction.
 - Over-forgetting**: unclamped entropy⁹ keeps pushing tokens that are already forgotten.

BLADE at a glance

Unlearning as a **constrained bilevel problem**: forget as much as possible while the retain loss stays within budget ε . The constraint residual is $r = \mathcal{L}_{CE}(\theta; \mathcal{B}'_{ret}) - \varepsilon$ on a fresh retain batch, so $r > 0$ means the budget is violated.

$$\min_{\theta} \mathcal{L}_{fgt}(\theta^*; \mathcal{D}_{forget}) \quad \text{s.t.} \quad \mathcal{L}_{CE}(\theta^*; \mathcal{D}_{retain}) \leq \varepsilon, \quad \theta^* \approx \arg \min_{\theta} \mathcal{L}_{CE}(\theta; \mathcal{D}_{retain})$$



1 Repair-first bilevel structure

- Inner loop = retain repair**: K fast SGD steps restore retain damage *before* each forget step.
- Outer loop = one small forget step**, sized so the next inner loop can fully repair it ($\eta_{in} > \eta_{out}$).
- Retain in the inner loop** (contrary to SCRUB,⁴ BLUR⁶): forgetting in the unconstrained inner loop would give K unopposed destructive steps per outer step.

2 Augmented Lagrangian with asymmetric dual update

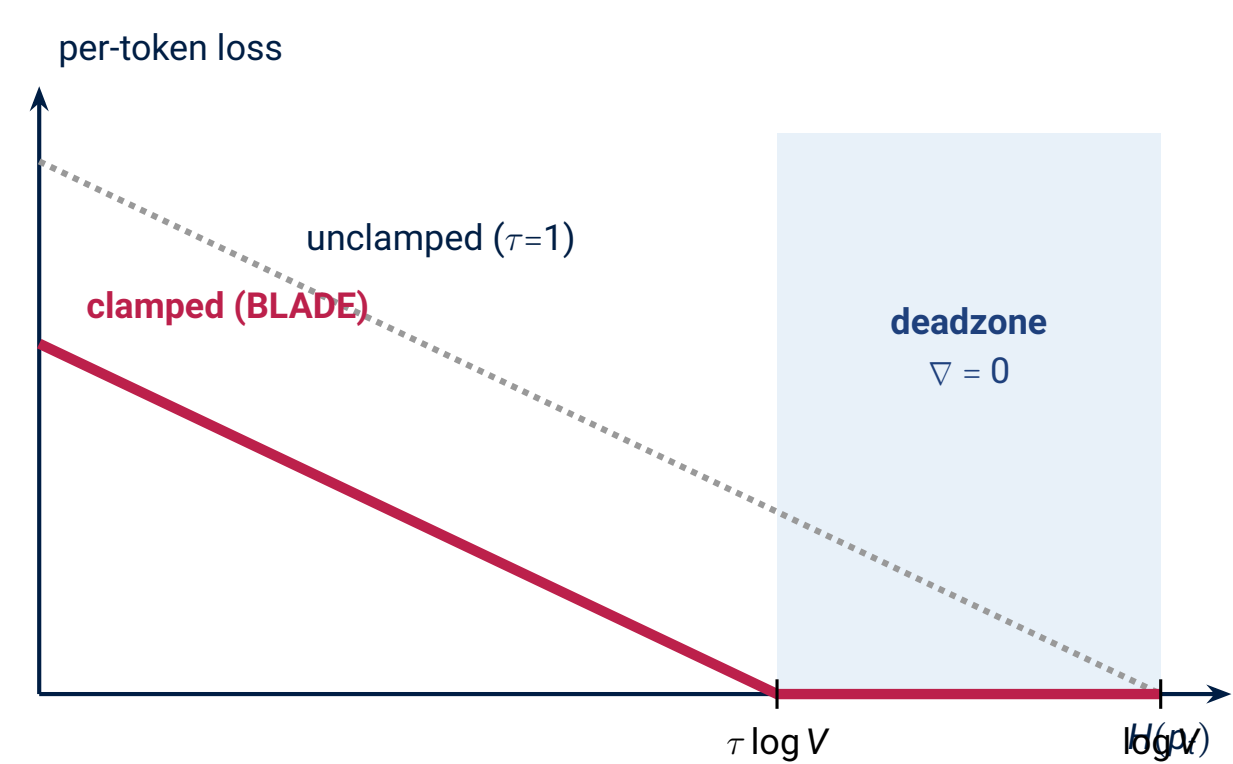
$$\mathcal{L}_{outer} = \mathcal{L}_{fgt} + \lambda r + \frac{\rho}{2} [\max(0, r)]^2 \quad \lambda \leftarrow \begin{cases} \lambda + \rho |r| & r > 0 \\ \lambda - \alpha \rho |r| & r \leq 0, \alpha \ll 1 \end{cases}$$

- $r = \mathcal{L}_{ret} - \varepsilon$ is measured on a fresh retain batch, and ε is set relative to the model's own retain loss (no manual calibration).
- An **inequality** constraint: there is no penalty while retain loss stays below ε .
- Ratchet**: λ jumps after a violation and decays only slowly. Symmetric updates (PDU¹) oscillate instead.

3 Clamped entropy forget loss

$$\mathcal{L}_{fgt} = \frac{1}{N} \sum_{t=1}^N \max(0, \tau \log V - H(p_t))$$

- Bounded**: $\mathcal{L}_{fgt} \in [0, \tau \log V]$.
- Self-stabilising**: the gradient is *exactly* zero once a token is forgotten.
- Reference-free**: no frozen model, unlike NPO.
- Vocabulary-invariant** τ (32K or 128K tokens).



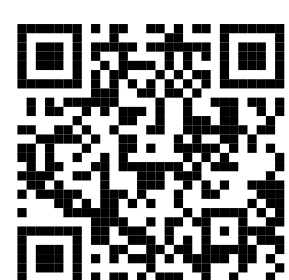
LoRA as a structural constraint: adapters on all projections, $W' = W_0 + \frac{\alpha}{\tau} BA$. The low-rank bottleneck limits weight drift, and disabling the adapters restores the original model.

Experimental setup

- TOFU**:⁵ Synthetic authors, Llama-3.2-1B/3B-Instruct, forget 1/5/10%.
- MUSE**:⁷ Books (Harry Potter) and News (BBC), Llama-2-7B.
- KnowUndo**:⁸ Copyright and privacy, LoRA-fine-tuned Llama-2-7B-chat.
- Baselines** (OpenUnlearning): GradAscent, GradDiff, NPO, SimNPO,² RMU, BLURNPO, PDU.
- Metrics**: per-benchmark harmonic mean (HM) of forget and retain axes, plus an LLM-judge composite HM_J for semantic leakage. All results are averaged over 5 seeds.

Takeaways and limitations

- A **bounded** forget loss, an **adaptive** retain constraint and a **low-rank** parameter space combine into a self-correcting unlearner.
- Gains hold across benchmarks, **under scaling** and **under repeated application**.
- Smooth training dynamics**, and extremely robust to hyperparameter choice.
- Adversarially robust** against black-box attacks.
- Limitations**: Vulnerable to re-learning attacks. Not yet tested on concept-level unlearning or in multimodal settings.



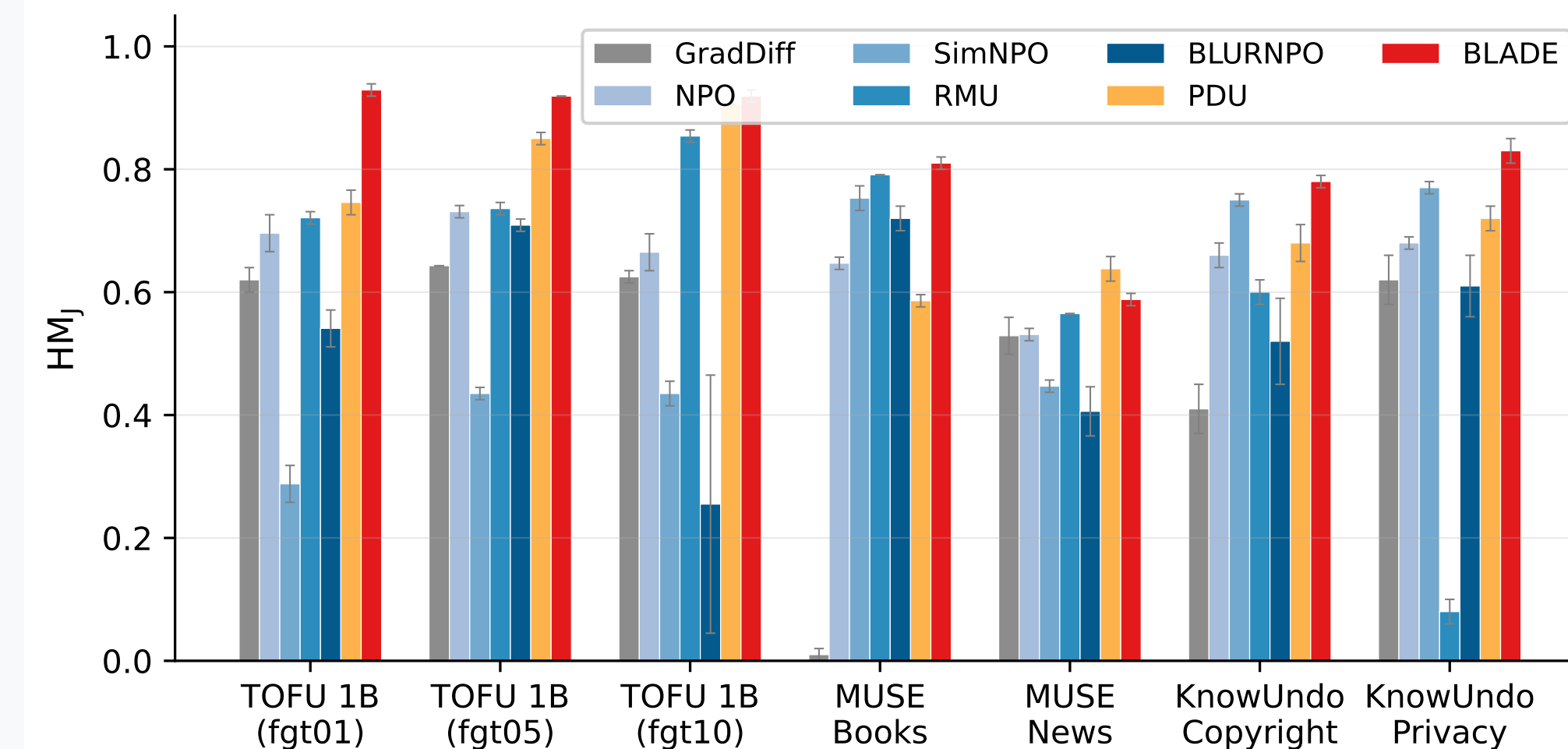
Toufikuzzaman, M., Mousavi, A., & Lee, D. (2026). BLADE: Bilevel Low-rank Augmented-Lagrangian Erasure for LLM Unlearning. To appear in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. arXiv:2608.22557

Code & configs: <https://github.com/tzpranto/blade>

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Finding 1. BLADE dominates across three benchmark families.

Benchmark (HM↑)	Best baseline	BLADE
TOFU 1B	fgt01 .690 PDU	.808
	fgt05 .740 PDU	.800
	fgt10 .797 PDU	.803
TOFU 3B	fgt01 .742 PDU	.846
	fgt05 .843 PDU	.842
	fgt10 .857 PDU	.836
MUSE	Books .753 SimNPO	.818
	News .577 PDU	.544
KnowUndo	Copyright .461 SimNPO	.477
	Privacy .546 PDU	.605

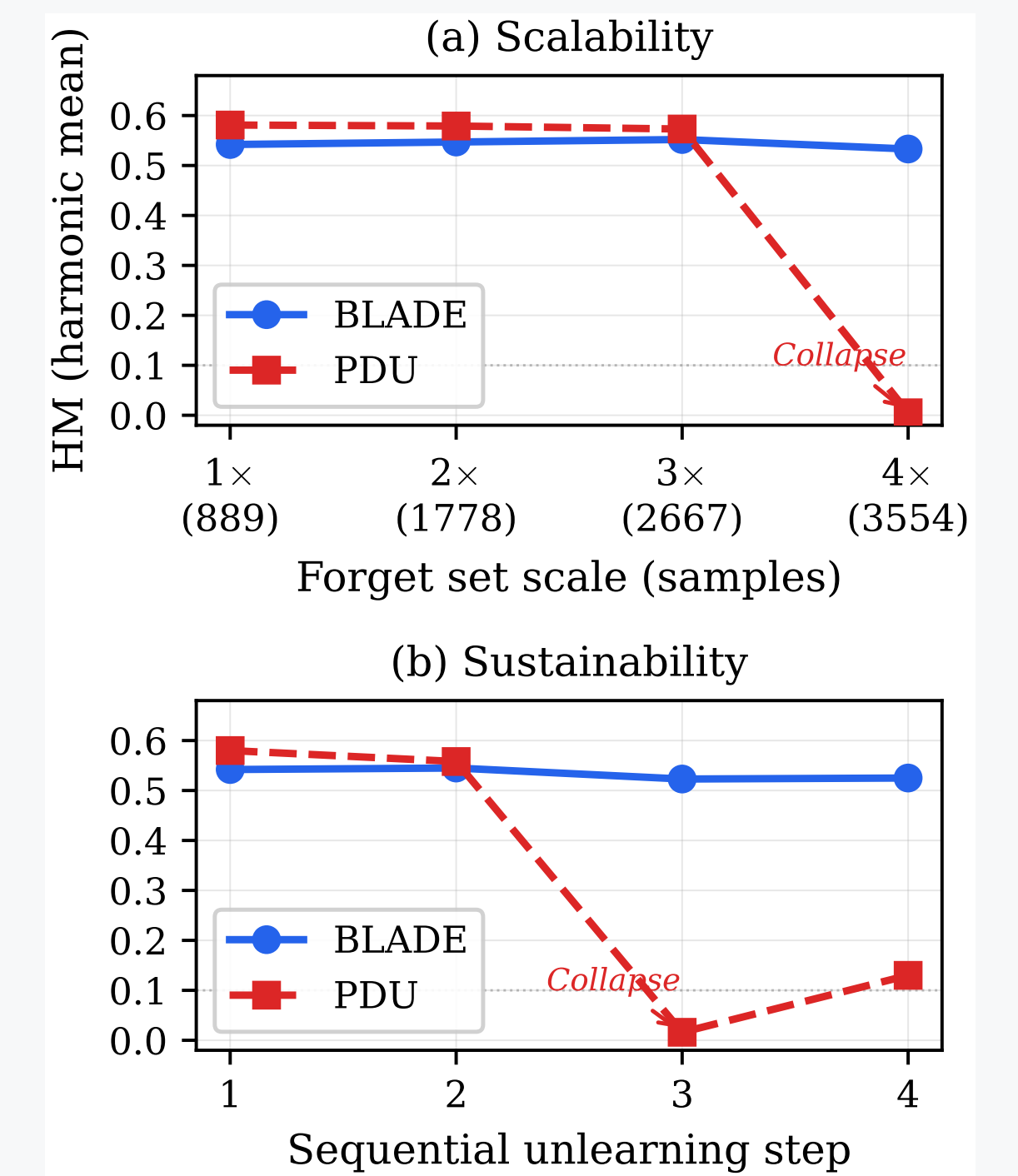


LLM-judge composite HM_J (semantic leakage, retain accuracy, response quality).
BLADE is highest on **6 of 7** settings.

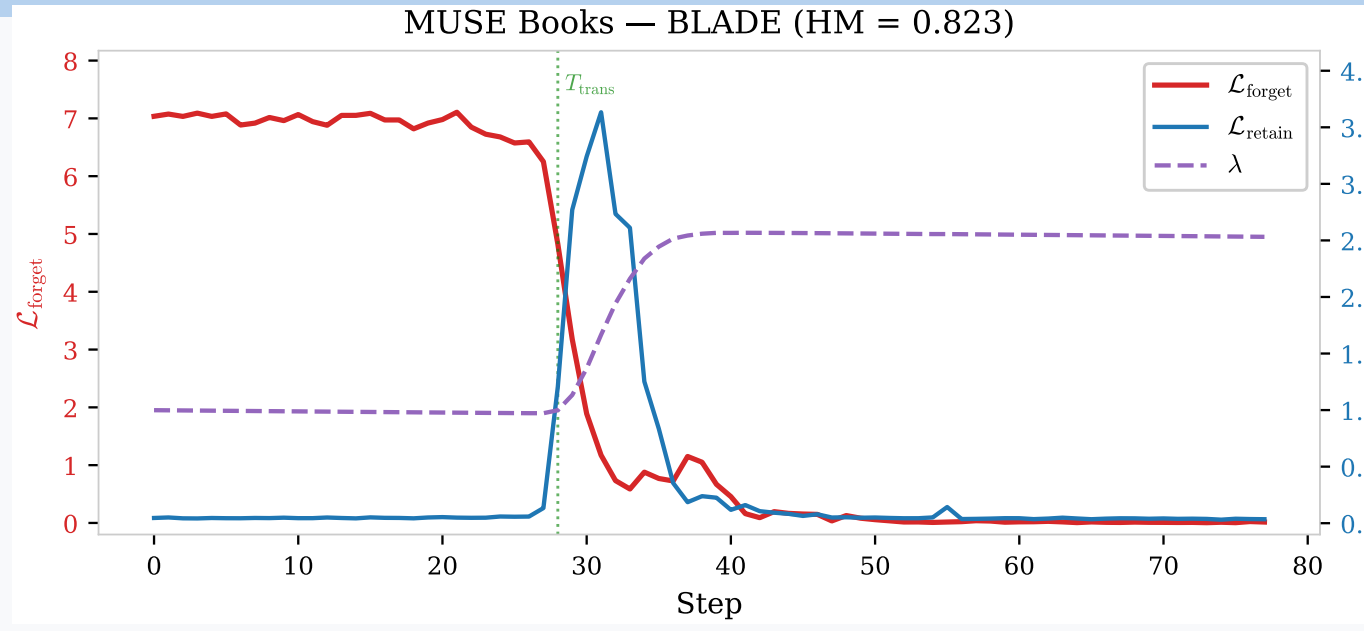
Finding 2. Robust under stress, where the strongest baseline collapses.

- MUSE News, head-to-head against PDU, the strongest News baseline.
- Scalability**: forget set grown up to 4×. PDU collapses at 4× ($HM = 0.005$).
- Sustainability**: sequential unlearning requests. PDU drops from 0.58 to 0.02 after only 2 steps.
- BLADE** stays within **HM 0.523–0.552** under every condition.

Bounded loss, the LoRA bottleneck and the ALM ratchet together prevent the model collapse at scale and under repeated application.



Finding 3. Smooth, self-correcting three-phase dynamics.



BLADE: warm-up → spike-and-ratchet → convergence

- Warm-up**: the forget loss falls while retain stays within ε .
- Spike-and-ratchet**: a transient retain violation permanently raises λ .
- Smooth convergence**: both losses decrease monotonically under the elevated λ .

The same pattern appears on every benchmark, model scale and dataset type, which makes the forget–retain trade-off interpretable and controllable.

Finding 4. BLADE's components complement each other.

MUSE Books	rk↑	HM↑	HM_J ↑
Full BLADE	.658	.823	.814
ALM off	.092	.233	.117
No LoRA (full FT)	.306	.459	.550
Swapped bilevel	.651	.506	.655
Unclamped ($\tau=1$)	.604	.780	.785
GA forget loss	.060	.160	.093
NPO forget loss	.662	.009	.495
Logit-margin loss	.000	.000	.000

- ALM** is the most critical piece: without it, retain collapses.
- LoRA** prevents representational drift (full fine-tuning: rk .658 → .306).
- Swapping the loops** makes λ diverge, which confirms the repair-first principle.
- Only a **bounded, self-saturating** forget loss is compatible with the constrained bilevel framework.

Finding 5. Resilient to black-box attacks and memory-efficient.

- Extraction**, using paraphrased or perturbed queries: best adversarial HM on TOFU 1B on every split.
- Jailbreak**: lowest attack success rate on every split (0.025 vs. PDU 0.087 on fgt01).
- Membership inference**: the only method whose PrivLeak score is positive on every split (no membership leakage).
- Compute**: 2–5× lower peak GPU memory than every baseline, for 1.5–2× the wall-clock time.

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